

COMSATS University Islamabad, Wah Campus



Software Requirement Design Specification

AutoFix: Intelligent Multi-Branch Vehicle Service Hub

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1 Introduction

1.1 System Introduction

AutoFix is a web-based platform designed for multi-branch vehicle service centers operating across cities such as Taxila, Wah Cantt, Hassanabdal, and Rawalpindi. The system provides a comprehensive solution for managing bookings, services, and spare parts efficiently. Customers can book appointments in real-time, view service durations and prices, interact with an AI assistant for guidance, and provide feedback. By entering vehicle details such as mileage and fuel efficiency, users receive AI-driven predictions using a Random Forest Classifier to determine maintenance needs and specific service requirements. Managers oversee branch operations, track mechanic workload, manage inventories, and generate analytical reports. AutoFix combines intelligent automation, predictive analytics, and centralized operations to improve customer experience and operational efficiency.

1.2 Background of the System

In an era where vehicle service centers struggle with operational inefficiencies, AutoFix was conceived to address critical challenges facing multi-branch automotive service providers. Traditional centers rely on manual processes and disconnected systems for managing appointments, tracking inventory, and allocating resources. This fragmentation leads to customer frustration through unclear pricing, lack of real-time updates, and inability to receive personalized maintenance guidance. Customers often struggle to identify exact maintenance needs, relying on guesswork rather than data-driven insights.

AutoFix's mission is to transform multi-branch service operations through digital innovation. By leveraging AI-powered predictive maintenance using machine learning algorithms, centralized inventory management, and intelligent resource allocation, AutoFix aims to provide a modern solution that addresses these challenges and empowers service centers to operate more efficiently and profitably.

1.3 Objectives of the System

- Centralized web platform for streamlined vehicle service operations across multiple branches
- AI-powered recommendation engine to predict maintenance needs based on vehicle parameters
- Real-time booking system with calendar integration and slot availability checking
- Comprehensive multi-branch inventory management with spare parts tracking and automated low-stock notifications
- Service tracking system with history logging, quality monitoring, and intelligent task allocation
- Centralized analytics dashboard with real-time insights on bookings, inventory, and revenue
- Notification and activity monitoring for instant updates on bookings and services
- Resource for vehicle owners with transparent pricing, service estimates, and educational content

1.4 Significance of the System

The significance of AutoFix lies in its comprehensive approach to solving operational challenges through digital transformation and artificial intelligence. By integrating booking management, AI powered maintenance prediction, inventory tracking, and analytics, AutoFix addresses inefficiencies in traditional service operations, transforming reactive service into proactive care and helping customers save costs and improve vehicle reliability, while also providing managers with real-time visibility and insights to inform data-driven decisions, ultimately creating a competitive advantage and positioning service centers for sustainable growth in the digital automotive service industry.

Application Area

- Multi-Branch Vehicle Service Centers
- Fleet Management Companies
- Automotive Workshops
- Dealership Service Departments

2 Overall Description

2.1 Product Perspective

AutoFix operates as a centralized web-based platform that connects customers, managers, and administrators through their respective browsers. It integrates seamlessly with backend components such as a MySQL database, Redis cache, AI model service, and communication modules including SMTP email and SMS gateways. The system functions as an interconnected ecosystem where the web application serves as the core controller for all operations. External users interact with the platform through a unified interface, while internal services ensure data processing, notifications, and intelligent recommendations. Overall, the platform is designed to provide a scalable, automated, and reliable environment for managing automotive service workflows.

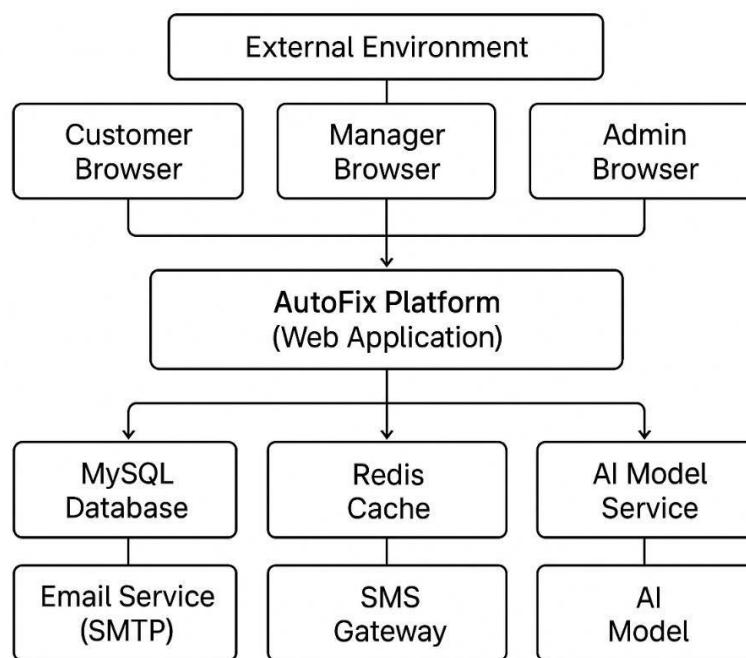


Figure 1 : General Diagram

2.2 Product Scope

AutoFix is a web-based platform designed for multi-branch vehicle service centers, offering a comprehensive solution for customers, managers, and administrators. The platform enables intelligent booking with real-time availability and calendar integration, AI-powered maintenance prediction, and multi-branch inventory management. Key features include JWT-based authentication, role-based access, profile management, service tracking, and centralized analytics with real-time insights on bookings, inventory, and revenue. The platform also includes integrated notifications, feedback module, and educational blogs, providing a unified solution for vehicle service management, aiming to enhance customer experience, streamline operations, and drive business growth

2.3 Product Functionality

- User Authentication & Profile Management
- Service Booking & Appointment System
- AI Customer Assistant & Recommendation Engine
- Multi-Branch Service & Inventory Management
- Vehicle Service Tracking & Allocation
- Centralized Analytics & Multi-Dashboard System
- Notification & Activity Monitoring System
- Customer Feedback & Blogs

2.4 Users and Characteristics

User Role	Primary Domain	Core Responsibilities
Customer	Service Seeker / Vehicle Owner	Register & manage personal profiles (with multiple vehicles), book service appointments, view service pricing, receive AI-based maintenance recommendations, track service status, and provide feedback/ratings.
Manager	Branch Operations / Local Oversight	Manage branch profiles and services, track mechanic workload and allocate tasks, monitor multi-branch inventory, view analytics & reports, and handle customer complaints.
Administrator	System-Wide Control / Configuration	Manage user accounts and role assignments, configure system settings and business rules, oversee all branches and performance metrics, and manage content (blogs, notifications).

2.5 Operating Environment

AutoFix is a web-based application accessible through modern browsers, designed to support multibranch vehicle service operations with centralized backend infrastructure.

2.5.1 Hardware Platform

The application operates on standard computing devices including desktop computers, laptops, tablets, and smartphones for field access.

2.5.2 Minimum Requirements

- **Processor:** Dual-core 2.0 GHz or equivalent
- **RAM:** At least 4 GB
- **Storage:** Minimum 100 MB for cached data
- **Display:** 1024x768 resolution minimum

2.5.3 Operating System and Version

- **Web Browsers:** Chrome 90+, Firefox 88+, Safari 14+, Edge 90+
- **Operating Systems:** Windows 10+, macOS 10.14+, Linux (Ubuntu 18.04+), Android
- **Internet Connection:** Stable broadband connection required for real-time features

3 Specific Requirements

3.1 Functional Requirements

3.1.1 User Authentication & Profile Management

Requirement	Description
RFR-001	Users must create accounts using email and password with JWT-based authentication.
RFR-002	Users shall log in with registered credentials and receive secure access tokens.
RFR-003	Password recovery through secure email verification for forgotten credentials.
RFR-004	Users can update profile details including name, contact info, and profile picture.
RFR-005	Customers can add multiple vehicles with make, model, year, mileage, and VIN.

3.1.2 Service Booking & Appointment System

Requirement	Description
RFR-006	Customers can view available time slots across branches.
RFR-007	System validates slot availability and branch working hours before confirming bookings.
RFR-008	Interactive calendar displays bookings and real-time availability.
RFR-009	Customers receive instant booking confirmation with service details.
RFR-010	Dashboard shows upcoming appointments, past services, and current status.

3.1.3 AI Customer Assistant & Recommendation Engine

Requirement	Description
RFR-011	Users enter vehicle parameters including mileage, fuel efficiency, and component conditions.
RFR-012	System identifies specific services required based on vehicle condition.
RFR-013	Chatbot provides personalized maintenance recommendations and cost estimates.

3.1.4 Multi-Branch Service & Inventory Management

Requirement	Description
RFR-014	Managers can create and update branch profiles with location and services offered.
RFR-015	Tracks staff allocation and their workload.
RFR-016	Inventory tracking for spare parts across all locations.
RFR-017	Automated low-stock alerts when inventory falls below thresholds.
RFR-018	System displays available services and pricing per branch.

3.1.5 Vehicle Service Tracking & Allocation

Requirement	Description
RFR-019	Complete service history logged for vehicles.
RFR-020	System tracks mechanic workload and current assignments.
RFR-021	Task allocation assigns services based on expertise and availability.
RFR-022	Managers can manually reassign tasks when needed.
RFR-023	Service status updates visible to customers.

3.1.6 Centralized Analytics & Multi-Dashboard System

Requirement	Description
RFR-024	Real-time analytics on bookings, revenue, and inventory across branches.
RFR-025	Branch-wise KPI dashboards showing performance metrics.
RFR-026	Interactive data visualizations including charts and graphs.
RFR-027	Exportable reports in PDF and Excel formats.
RFR-028	Comparison views between branches and time periods.

3.1.7 Notification & Activity Monitoring System

Requirement	Description
RFR-029	Real-time service status updates sent to customers.
RFR-030	Inventory alerts sent to managers for low-stock items.
RFR-031	Activity logs track all system actions with timestamps.
RFR-032	Users can configure notification preferences.

3.1.8 Customer Feedback & Blogs

Requirement	Description
RFR-033	Customers can submit ratings and reviews after service completion.
RFR-034	Testimonials module displays positive feedback.
RFR-035	Blog system for publishing car care tips and service-related articles.

3.2 Behaviour Requirements

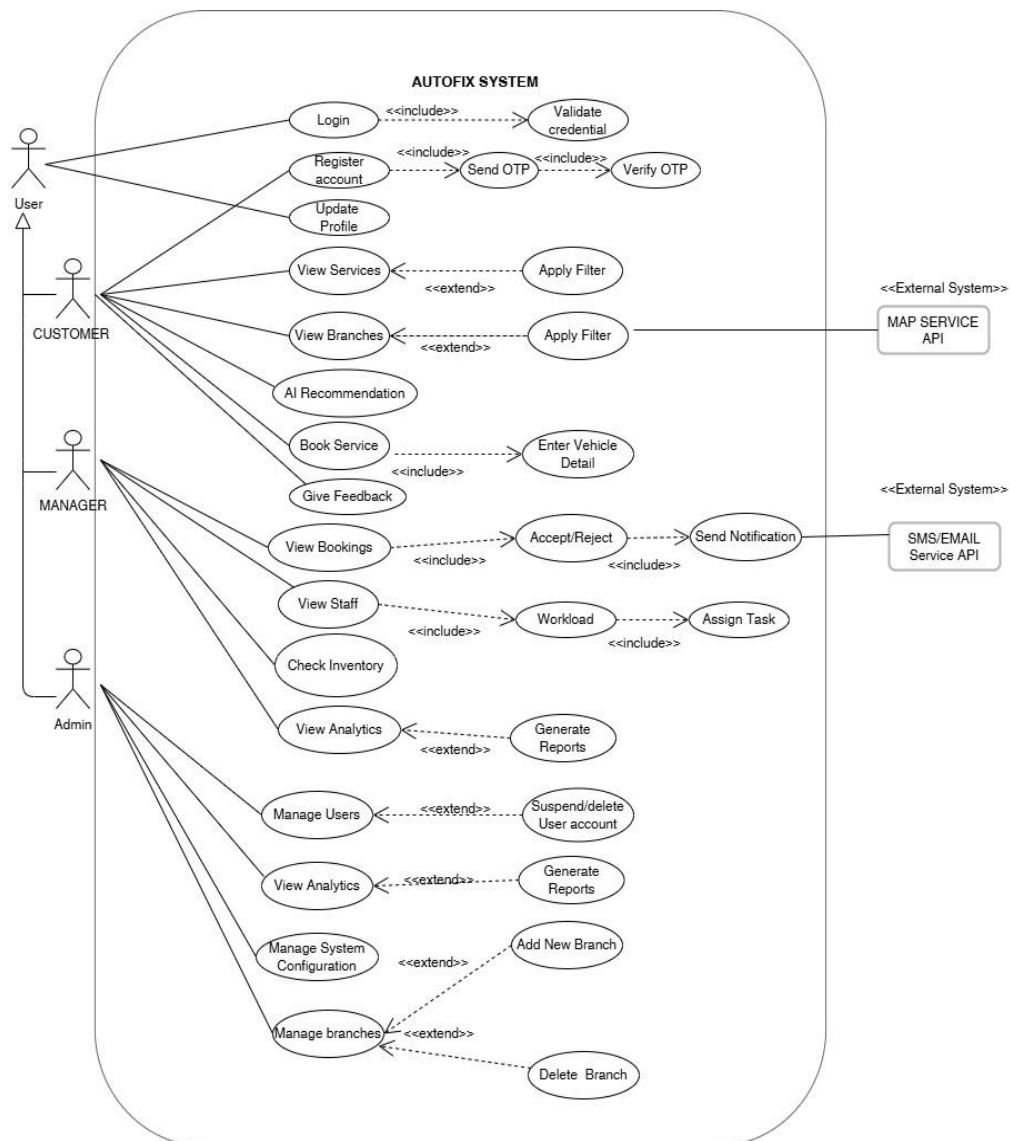


Figure 2 : Use Case Diagram

3.2.1 Use Case Tables for AutoFix System

Table 1: Login – Use Case Description

Attributes	Detail
Use Case Name	Login
Actors	Customer, Manager, Admin
Description	Allows a registered user to authenticate and access the system by validating credentials.
Pre-condition	The user must have a registered account in the system.
Steps	1. User navigates to the Login screen. 2. User enters email/username and password. 3. System validates credentials against the database. 4. If valid, system grants access to the role-specific dashboard (Customer/Manager/Admin).
Post-condition	User is successfully logged in and redirected to their respective dashboard.

Table 2: Register Account – Use Case Description

Attributes	Detail
Use Case Name	Register Account
Actors	Customer
Description	Allows a new user to create an account by verifying their email or phone number using OTP.
Pre-condition	The user is not already registered in the system.
Steps	1. User selects "Register" and enters personal details (Name, Email, Phone). 2. System generates and sends OTP to the user's email/phone. 3. User enters the received OTP. 4. System verifies the OTP. 5. Account is created successfully.
Post-condition	User account is created and user is redirected to the login screen or dashboard.

Table 3: Update Profile – Use Case Description

Attributes	Detail
Use Case Name	Update Profile
Actors	Customer, Manager, Admin
Description	Allows any registered user to modify their personal information and account settings.
Pre-condition	The user must be logged in to the system.
Steps	1. User navigates to "My Profile" section. 2. User selects "Edit Profile". 3. User modifies details (Name, Email, Phone Number, Profile Picture, Password). 4. User clicks "Save Changes". 5. System validates and updates the record in the database.
Post-condition	User's profile information is successfully updated and saved.

Table 4: View Services – Use Case Description

Attributes	Detail
Use Case Name	View Services
Actors	Customer
Secondary Actor	MAP SERVICE (External System)
Description	Allows customers to browse available auto repair services with filtering options.
Pre-condition	Customer must be logged in; internet connection required.
Steps	1. Customer navigates to "View Services" section. 2. System retrieves and displays all available services. 3. Customer applies filters (Service Type, Location, Price Range). 4. System interacts with MAP SERVICE to show service locations. 5. Customer views filtered results.
Post-condition	Customer views a list of services matching their search criteria.

Table 5: View Branches – Use Case Description

Attributes	Detail
Use Case Name	View Branches
Actors	Customer
Description	Allows customers to view all auto repair branches with location and contact details.
Pre-condition	Customer must be logged in.
Steps	1. Customer navigates to "View Branches" section. 2. Customer applies filters (Location, City, Availability). 3. System retrieves and displays branch information. 4. System may use MAP SERVICE to show branch locations on a map.
Post-condition	Customer views a list of branches with details.

Table 6: AI Recommendation – Use Case Description

Attributes	Detail
Use Case Name	AI Recommendation
Actors	Customer
Description	Provides AI-generated service recommendations based on vehicle condition and service history.
Pre-condition	Customer must be logged in and have vehicle details registered.
Steps	1. Customer navigates to "AI Recommendation" feature. 2. System analyzes customer vehicle data and history. 3. AI generates personalized recommendations. 4. System displays recommended services with estimated cost and urgency.
Post-condition	Customer receives tailored service recommendatio

Table 7: Book Service – Use Case Description

Attributes	Detail
Use Case Name	Book Service
Actors	Customer
Secondary Actor	SMS/EMAIL Service (External System)
Description	Allows customers to book an auto repair service by entering vehicle details and selecting a preferred time slot.
Pre-condition	Customer must be logged in and have selected a service.
Steps	1. Customer selects a service. 2. Enters vehicle details (Make, Model, Registration Number). 3. Chooses date and time slot. 4. Confirms booking. 5. System sends booking confirmation via SMS/EMAIL. 6. System saves booking details.
Post-condition	Booking is confirmed and customer receives notification.

Table 8: Give Feedback – Use Case Description

Attributes	Detail
Use Case Name	Give Feedback
Actors	Customer
Description	Allows customers to provide ratings and feedback after receiving a service.
Pre-condition	Customer must have completed at least one service booking.
Steps	1. Customer navigates to “Give Feedback”. 2. Selects a completed booking. 3. Rates service (1–5 stars) and writes comments. 4. Submits feedback. 5. System stores feedback.
Post-condition	Feedback is recorded and accessible to managers/admin.

Table 9: View Bookings – Use Case Description

Attributes	Detail
Use Case Name	View Bookings
Actors	Manager
Secondary Actor	SMS/EMAIL Service
Description	Allows managers to view customer bookings and manage their status.
Pre-condition	Manager must be logged in.
Steps	1. Manager navigates to “View Bookings”. 2. System retrieves all bookings. 3. Manager accepts/rejects bookings. 4. If accepted, system sends notification. 5. System updates booking status.

Attributes	Detail
Post-condition	Booking status is updated and customer notified.

Table 10: View Staff – Use Case Description

Attributes	Detail
Use Case Name	View Staff
Actors	Manager
Description	Allows managers to view staff, check availability, and assign tasks.
Pre-condition	Manager must be logged in.
Steps	1. Manager navigates to “View Staff”. 2. System shows staff list with status (Available/Busy). 3. Manager assigns tasks. 4. System updates workload and assignments.
Post-condition	Staff tasks are assigned and database updated.

Table 11: Check Inventory – Use Case Description

Attributes	Detail
Use Case Name	Check Inventory
Actors	Manager
Description	Allows managers to monitor and manage inventory of spare parts.
Pre-condition	Manager must be logged in.
Steps	1. Manager goes to “Check Inventory”. 2. System retrieves inventory data. 3. Manager reviews low-stock items. 4. Manager updates stock levels or initiates restocking.
Post-condition	Inventory records are updated.

Table 12: Manage Users – Use Case Description

Attributes	Detail
Use Case Name	Manage Users
Actors	Admin
Description	Allows admins to suspend, delete, or modify user accounts.
Pre-condition	Admin must be logged in.
Steps	1. Admin navigates to “Manage Users”. 2. System displays all users. 3. Admin selects a user. 4. Admin suspends, deletes, or updates account. 5. System updates user status.
Post-condition	User account is modified or removed.

Table 13: View Analytics – Use Case Description

Attributes	Detail
Use Case Name	View Analytics
Actors	Admin
Description	Allows admins to view system performance metrics and reports.
Pre-condition	Admin must be logged in.
Steps	1. Admin navigates to “View Analytics”. 2. System generates reports. 3. Dashboard displays charts/graphs (Bookings, Revenue, Users, Trends).
Post-condition	Admin views analytics dashboard.

Table 14: Manage System Configuration – Use Case Description

Attributes	Detail
Use Case Name	Manage System Configuration
Actors	Admin
Description	Allows admins to configure system settings, update pricing, and modify parameters.
Pre-condition	Admin must be logged in with elevated privileges.
Steps	1. Admin enters “System Configuration”. 2. Admin adds new branches. 3. Admin updates pricing or service details. 4. System validates and saves changes.
Post-condition	Configuration changes are applied across the system.

Table 15: Manage Branches – Use Case Description

Attributes	Detail
Use Case Name	Manage Branches
Actors	Admin
Description	Allows admins to add, update, or delete branch locations.
Pre-condition	Admin must be logged in.
Steps	1. Admin navigates to “Manage Branches”. 2. Admin views all branches. 3. Admin adds, updates, or deletes branches. 4. System updates branch data.
Post-condition	Branch information is successfully modified.

3.3 External Interface Requirements

3.3.1 User Interfaces

Home Screen

The home screen provides a central dashboard with quick access to Booking, Services, the AI Assistant, user Profile, and Analytics. It includes a navigation bar, notification bell, and a user profile dropdown for smooth and intuitive navigation.

Booking Module Screen

The booking module features an interactive calendar that displays available time slots across branches. Users can select a branch, choose a service with pricing, and complete their appointment through a booking confirmation form.

AI Recommendation Screen

The AI screen allows users to input vehicle parameters to receive instant maintenance predictions. It displays recommended services and includes a chatbot interface for answering user queries.

Manager Dashboard

The manager dashboard presents branch performance metrics and mechanic workload visualizations. It also shows real-time inventory status across branches and provides quick action buttons for efficient task allocation.

Admin Panel

The admin panel contains tools for managing users, system settings, and branch-wide configurations. It includes analytics across all branches and a content management section for publishing blogs and updates.

3.3.2 Hardware Interfaces

- No specialized hardware required
- Standard web browsers on desktop and mobile devices
- Internet connectivity for cloud services
- Optional: Camera for vehicle photo uploads

3.3.3 Software Interfaces

Backend Services

- Django RESTful APIs for business logic
- MySQL database for structured data
- Redis for caching and session management

AI Model Integration

- Random Forest Classifier using Python's scikit-learn library
- Trained on historical vehicle maintenance data

External Services

- Email service (SMTP) for notifications
- SMS gateway for text alerts

Libraries and Frameworks

- React.js for frontend
- Chart.js for data visualization

3.3.4 Communications Interfaces

Network Protocols

- HTTPS for all client-server communication
- RESTful API architecture

Security

- Secure token-based authentication

Notifications

- Email via SMTP
- In-app push notifications
- SMS via third-party gateway

4 Other Non-functional Requirements

4.1 Performance Requirements

4.1.1 General Performance

- Response time under 2 seconds for standard operations
- Page load time under 3 seconds on broadband
- Support for 300 concurrent users

4.1.2 Booking Module

- Real-time slot availability updates within 1 second
- Booking confirmation processed within 2 seconds
- Calendar loads within 1.5 seconds

4.1.3 AI Prediction Module

- Maintenance prediction generated within 3 seconds
- Chatbot response time under 2 seconds

4.1.4 Data Synchronization

- Data consistency maintained across branches
- Automatic retry for failed synchronization

4.1.5 Analytics Module

- Dashboard data refreshed every 30 seconds
- Report generation under 5 seconds

4.1.6 Server Performance

- API response time under 500ms
- Database query optimization for complex joins
- Load balancing for high traffic periods

4.1.7 Resource Utilization

- Efficient memory usage with caching strategies
- Optimized database queries to reduce load
- CDN for static assets

4.1.8 Error Handling and Recovery

- Graceful degradation when services fail
- Automatic retry for transient errors
- User-friendly error messages with recovery options

4.2 Safety and Security Requirements

4.2.1 Safety Requirements

- Data backup every 24 hours
- Disaster recovery plan with RTO of 4 hours
- Data integrity checks on critical operations
- Audit logs for compliance

4.2.2 Security Requirements

- JWT-based authentication with token expiration
- Role-based access control (RBAC) for different user types

4.3 Software Quality Attributes

- Usability: User-friendly interface, intuitive navigation, and clear instructions
- Reliability: Robust error handling, fault tolerance, and high availability
- Performance: Fast response times, efficient data retrieval, and optimized database queries
- Security: Secure authentication, authorization, data encryption, and protection against common web attacks
- Scalability: Designed to handle increased traffic, data growth, and user expansion
- Maintainability: Modular architecture, clean code, and comprehensive documentation
- Flexibility: Adaptable to changing requirements, easy integration with third-party services
- Availability: High uptime, redundant systems, and disaster recovery mechanisms
- Compatibility: Cross-browser, cross-platform, and mobile-friendly design

5 Design Description

5.1 Composite Viewpoint

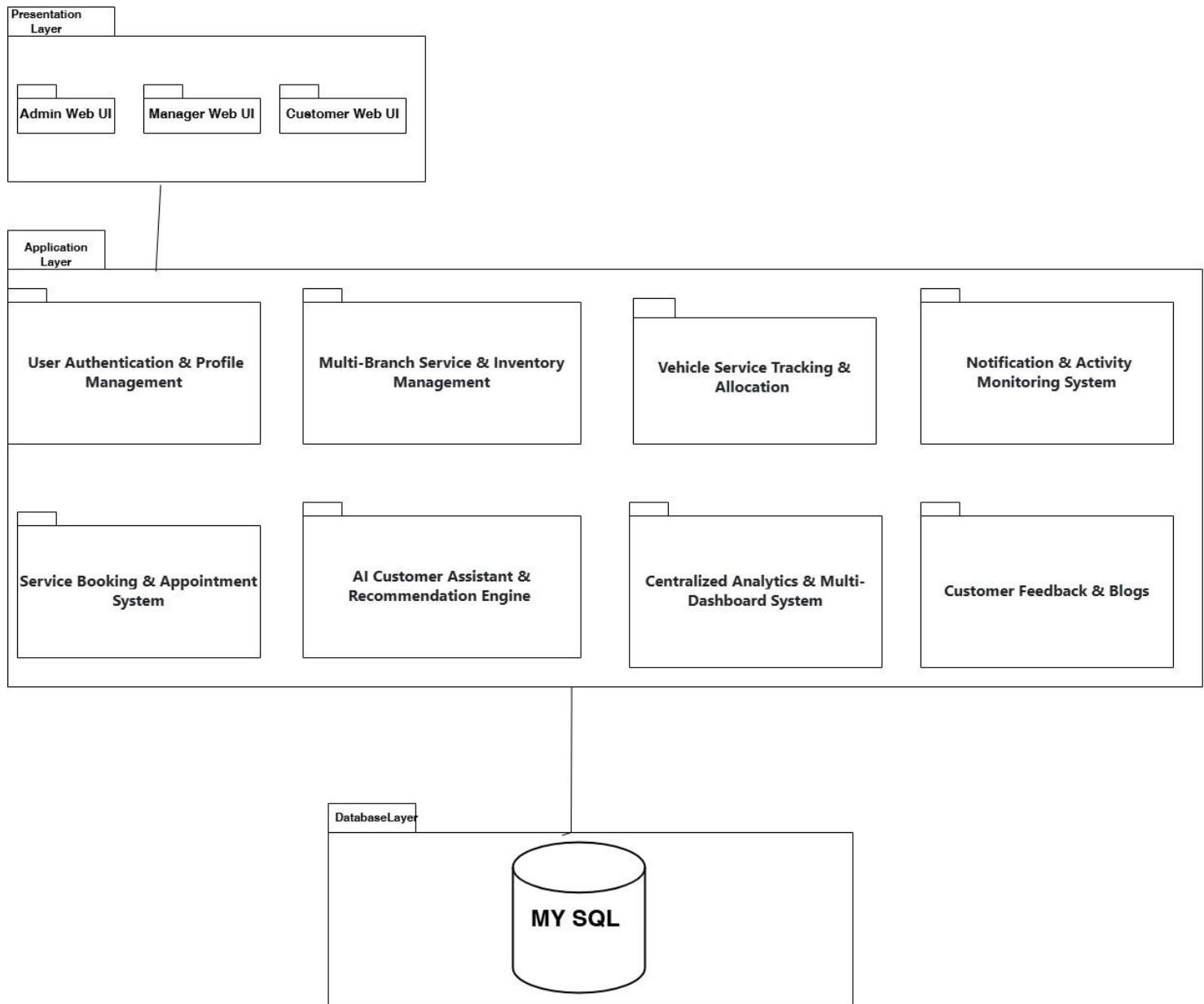


Figure 3 : Package Diagram

5.2 Logical Viewpoint

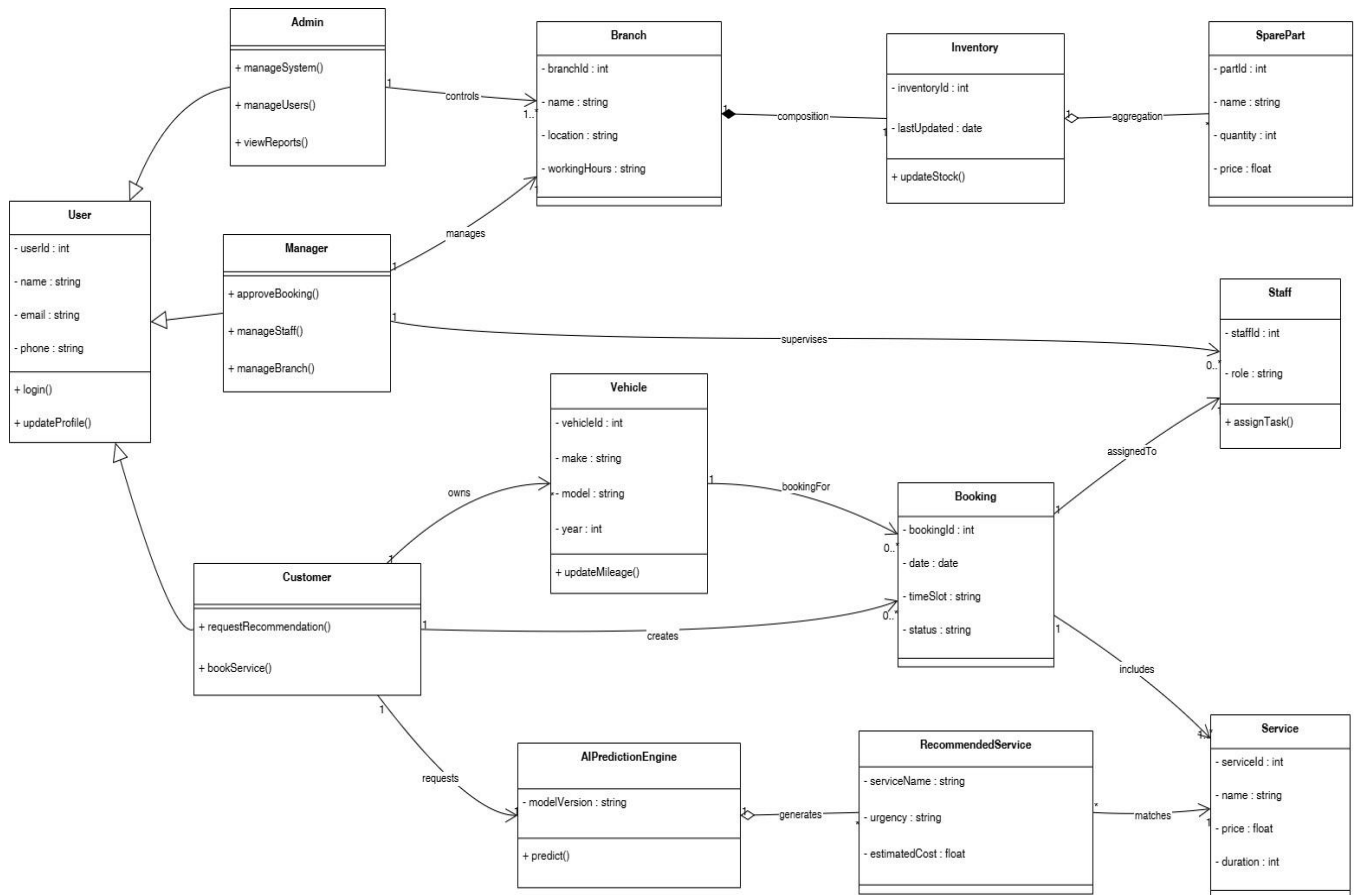


Figure 4: Class Diagram

5.3 Information Viewpoint

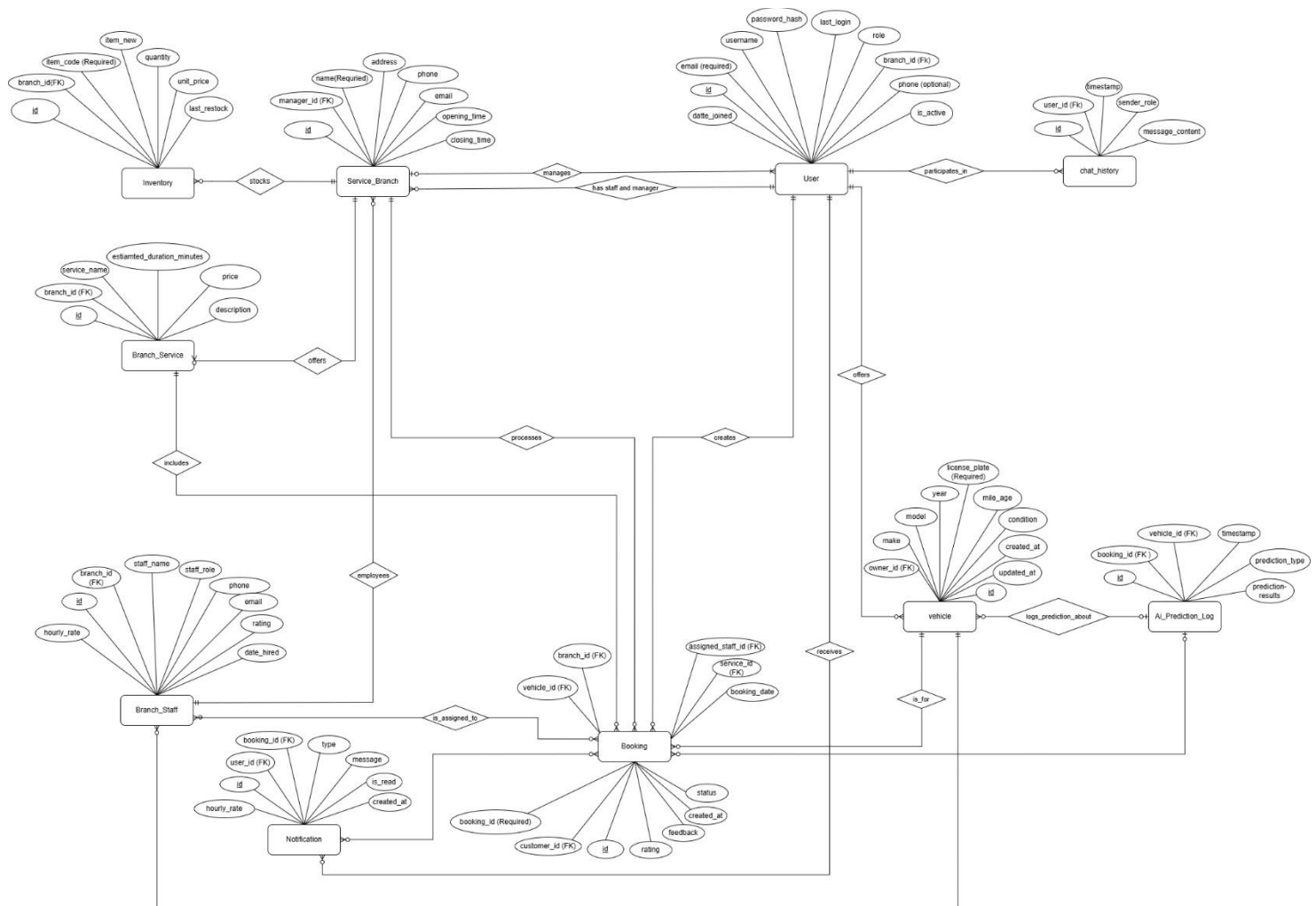


Figure 5 :ERD Diagram

5.4 Interaction Viewpoint

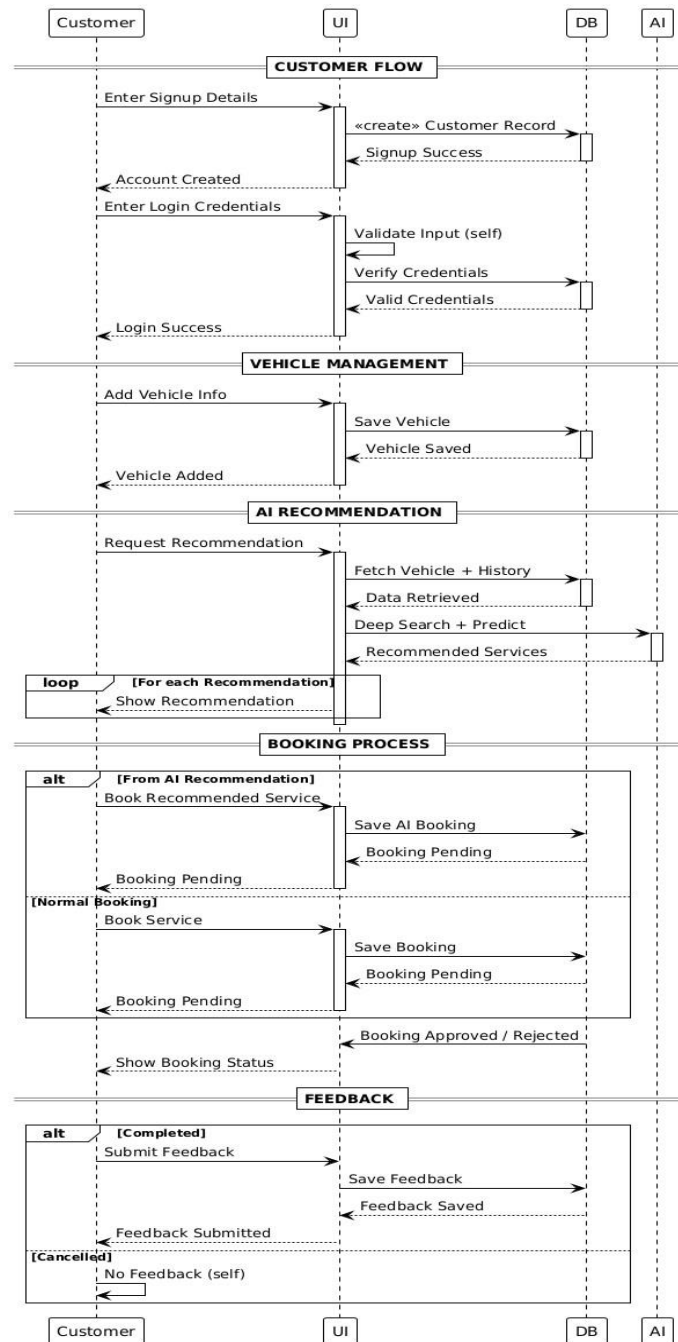


Figure 6 : Customer Sequence Diagram

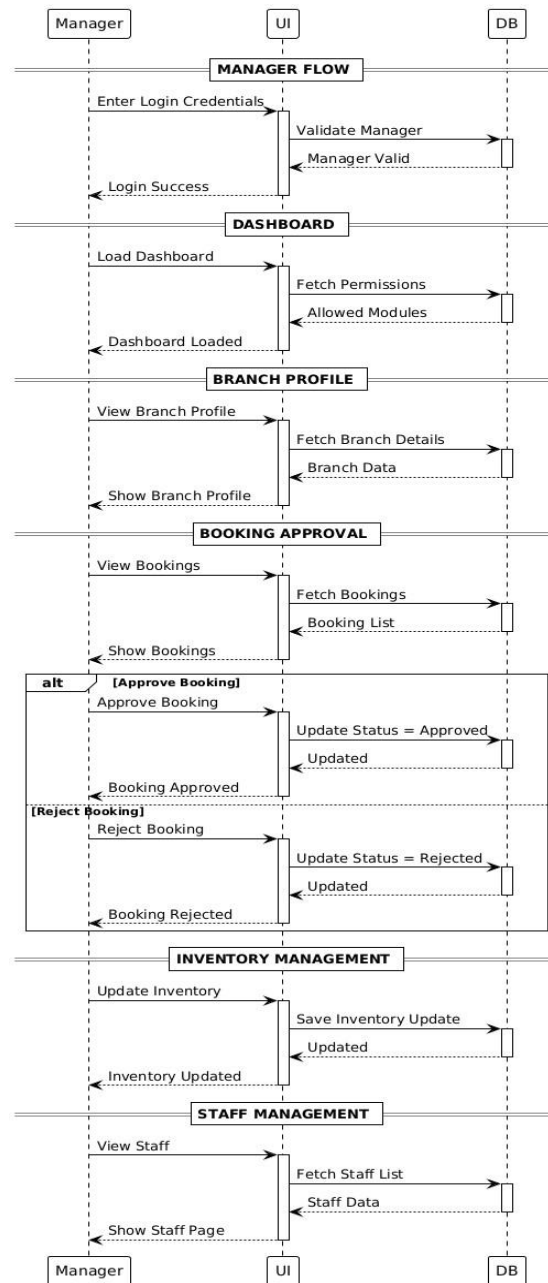


Figure 7 : Manager Sequence Diagram

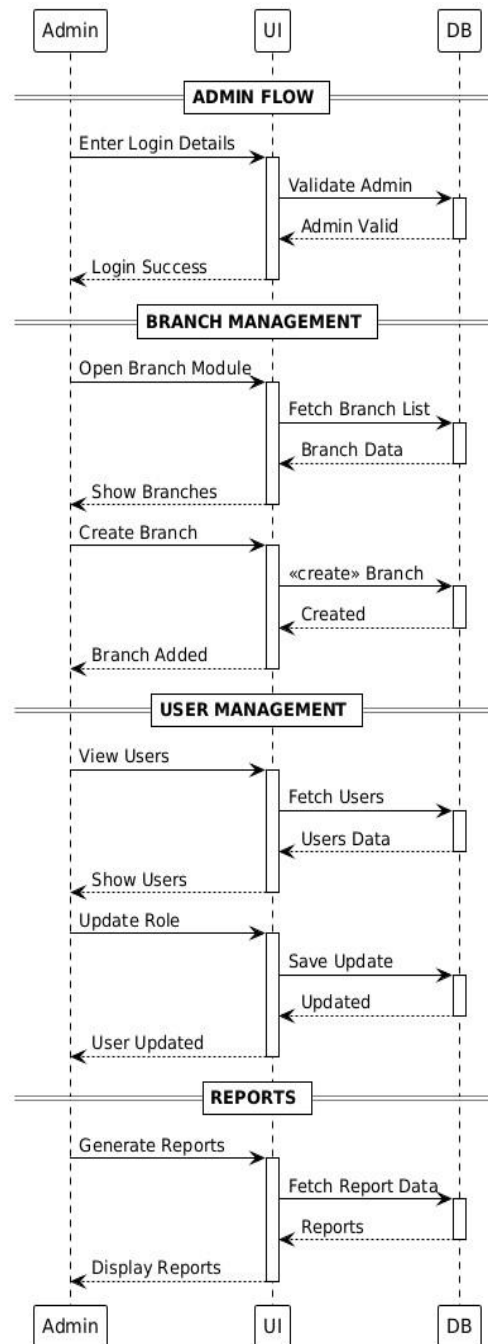


Figure 8 : Admin Sequence Diagram

5.5 State Dynamics Viewpoint

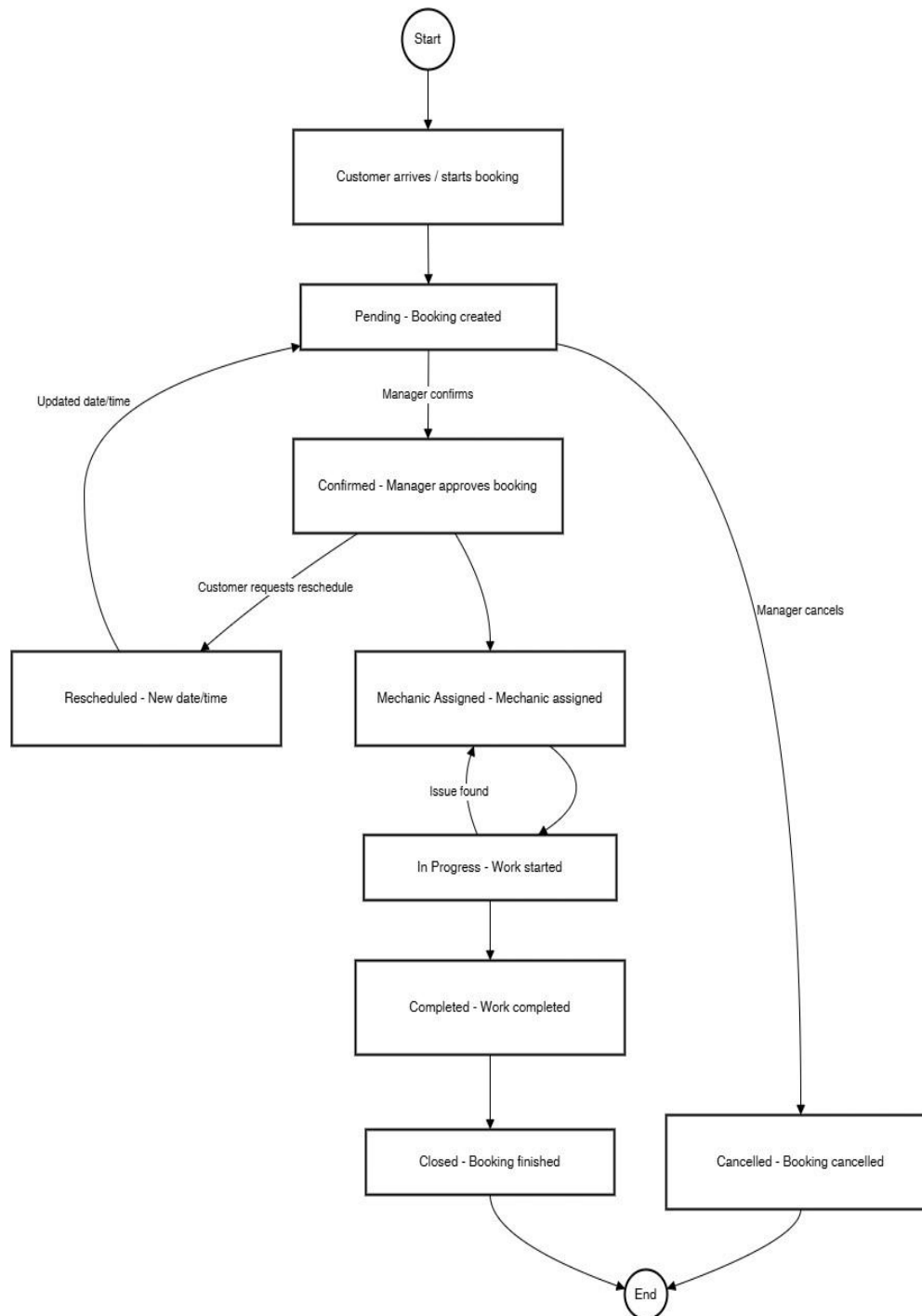


Figure 9 : Complete Booking Process State Diagram

5.6 Algorithm Viewpoint

5.6.1 AI Maintenance Prediction Algorithm

ALGORITHM PredictMaintenance(vehicle_data)

INPUT: mileage, fuel_efficiency, condition, age

OUTPUT: {maintenance_type, urgency, confidence}

BEGIN model ←

LOAD("random_forest_model.pkl")

features ← [

NORMALIZE(vehicle_data.mileage),

NORMALIZE(vehicle_data.fuel_efficiency),

ENCODE(vehicle_data.condition),

NORMALIZE(vehicle_data.age)

]

predictions ← []

FOR each tree IN model.trees DO

predicted_class ← tree.predict(features)

predictions.append(predicted_class)

END FOR

maintenance_type ← MAJORITY_VOTE(predictions) confidence ←

COUNT(predictions == maintenance_type) / TOTAL_TREES

urgency ← GET_URGENCY(vehicle_data, confidence)

RETURN {

maintenance_ty

pe, urgency,

confidence

}

END

5.6.2 Booking Creation Algorithm

ALGORITHM CreateBooking(customer_id, vehicle_id, booking_data)

INPUT: customer_id, vehicle_id, date, branch, time_slot

OUTPUT: booking_confirmation

BEGIN

VERIFY customer exists

VERIFY vehicle exists AND belongs to customer

prediction ← PredictMaintenance(vehicle_data) estimate ←

EstimateCostAndTime(prediction.maintenance_type, vehicle.age)

booking_id ← GENERATE_ID()

booking ←

{ id:

booking_id,

customer_id

, vehicle_id,

branch_id:

booking_da

ta.branch,

service_typ

e:

prediction.

maintenanc

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time_slot:

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es, status:

"PENDING

"

}

DATABASE.INSERT("bookings", booking)

SEND_NOTIFICATION("New booking created", customer_id)

RETURN booking

END